1. Scenario:

Your company is building a smart customer support chatbot to handle FAQs, ticket creation, and escalation.

Question:

Which Multi-Agent architecture would work best?

- Create a Main Coordinator Agent to manage the conversation flow.
- Add specialized agents such as FAQAgent, TicketAgent, and EscalationAgent.
- The coordinator receives the user question and identifies the intent.
- It forwards the request to the correct specialized agent.
- The agent processes the request and returns the response to the user.

2. Scenario:

In a Nested Agent setup, one Sub-Agent fails to complete its sub-task.

Question:

What should the Master Agent ideally do?

- The Master Agent detects that the sub-agent failed or returned an error.
- It logs the issue for monitoring or debugging.
- The Master Agent retries the task or assigns it to another sub-agent.
- If retries fail, the Master Agent handles the task itself or simplifies it.
- Finally, it responds to the user or escalates the problem.

3. Scenario:

You're building a conversational AI for a healthcare app. One module checks symptoms, and another suggests next steps.

Question:

How should the agent system be designed for smooth user interaction?

- Use a Conversation Manager Agent to interact with the user.
- Create a SymptomCheckerAgent to analyze symptoms.
- Add a RecommendationAgent to suggest next steps.
- The manager sends the user input to the symptom agent first.
- The results are passed to the recommendation agent and then shown to the user.

4. Scenario:

During real-time chat, the agent responds slowly and often delays the next message.

Question:

What could be the reason and how to solve it?

- Check if the LLM or API response time is slow.
- Reduce large prompts or unnecessary context sent to the model.
- Use faster models or caching for repeated answers.
- Process tasks asynchronously or in parallel when possible.
- Monitor system performance and optimize slow components.

5. Scenario:
An agent-based project for a smart assistant is expanding. You now need to add calendar handling, email reminders, and weather updates.
Question:
How will you structure your agent system?

- Create a Main Assistant Agent to handle user interaction.
- Add specialized agents like CalendarAgent, EmailAgent, and WeatherAgent.
- The main agent understands the user request.
- It sends the request to the appropriate specialized agent.
- The result is returned to the user through the main agent.

6. Scenario:
Your Conversational Agent often gives generic replies and doesn't adapt to user preferences over time.
Question:
What's missing in the agent's design?

- Store user profile and preferences in a memory system.
- Add a Memory Module to track previous interactions.
- Use the stored context when generating responses.
- Adapt replies based on user history and behavior.
- Continuously update user data to improve personalization.

7. Scenario:
Your chatbot is repeating the same answers even when the context changes.
Question:
What's possibly wrong and how to fix it?

- Check if the conversation history is being stored correctly.
- Ensure the agent receives updated context in each prompt.
- Remove repeated or irrelevant past messages.
- Use dynamic prompts based on current conversation state.
- Test with different scenarios to confirm correct context handling.

8. Scenario:
An AI agent is making decisions based on user financial data.
Question:
What design considerations should you follow?

- Protect financial data using encryption and secure storage.
- Implement strict access control and authentication.
- Ensure transparent decision logic for auditability.
- Follow privacy regulations and compliance rules.
- Provide users with clear explanations of decisions.

9. Scenario:
You're building a travel assistant agent that pulls flight, hotel, and weather data.
Question:
How will you manage multi-source interaction?

- Create a Coordinator Agent to manage requests.
- Add specialized agents like FlightAgent, HotelAgent, and WeatherAgent.
- Each agent connects to its respective data API.
- The coordinator collects responses from all agents.
- Combine the results into a single travel recommendation for the user.

10. Scenario:
Your chatbot was used for FAQs before. Now it's becoming a full personal assistant.
Question:
What changes will you make to the agent design?

- Move from a simple FAQ system to a multi-agent architecture.
- Add specialized agents for tasks like reminders, scheduling, and search.
- Implement a task planner agent to manage complex requests.
- Add memory and user context tracking for personalization.
- Integrate external tools and APIs to support advanced actions.